Självständigt arbete (examensarbete), 15 hp, för
Kandidatexamen i Informatik
Termin år: VT 2021
Fakulteten för Ekonomi

Högskolan Kristianstad
291 88 Kristianstad
044-250 30 00
www.hkr.se

Battling misinformation with a graphic interface

Exploring navigation through video to support digital vigilantists.

Victor Sandino & Anton Dahlström

Författare:
Victor Sandino & Anton Dahlström

Titel:
Att bestrida missinformation med ett grafiskt gränssnitt: Ett utforskande av videonavigation för att stödja digitala vigilantister

Handledare:
Montathar Faraon

Examinator:
Martin Wetterstrand

Sammanfattning

Digital vigilantism är ett fenomen som växer och blir mer och mer populärt (Chia, 2018; Yardley, Lynes, Wilson, & Kelly, 2016). Digitala vigilantister har under ett flertal tillfällen misslyckats i att korrekt identifiera både förövare och offer i händelser som omger socialt oönskade handlingar (Trottier, 2016; Yardley et al., 2016). Denna studie avser att använda konceptdriven designforskning (Stolterman & Wiberg, 2010) för att utveckla ett koncept som ska förbättra precisionen av utredningar genomförda av digitala vigilantister. Konceptet som presenteras i denna studie är en interaktiv videospelare som tillåter användare att navigera i videomaterial genom att interagera direkt med objekt i videorna. På detta vis ska känslan av position och riktning förhöjas. Resultat i denna studie utgörs av nya insikter av aspekter som behövs hållas i åtanke vid skapande av koncept för denna målgrupp. Detta inkluderar den komplexa etiska problametik som uppstår kring socialt oönskade handlingar och utredningar av dessa handlingar.

Ämnesord: Digital vigilantism, Design, Sociala media, videointeraktion

Authors:
Victor Sandino & Anton Dahlström

Title:
Battling misinformation with a graphic interface: Exploring navigation through video to support digital vigilantists.

Supervisor:
Montathar Faraon

Examiner:
Martin Wetterstrand

Abstract

Digital vigilantism is a phenomenon that is growing and gaining more popularity (Chia, 2018; Yardley et al., 2016). This practice has in various occasions failed to correctly identify both victims and perpetrators in events surrounding socially undesirable acts (Trottier, 2016; Yardley et al., 2016). This study aims to use Concept driven design research (Stolterman & Wiberg, 2010) to develop a concept that will enhance the precision of the investigative research performed by digital vigilantists. The concept presented in this study is an interactive video player that allows users to navigate through video by interacting directly with objects in the video and enhancing the sense of position and direction. This study results in new insights of aspects to take in mind when creating a concept for this particular target group, this includes the complex ethical aspects that arises when dealing with socially undesirable acts and the public investigation of these.

Keywords: Digital vigilantism, Design, Social media, Video interaction

Table of contents

1. INTRODUCTION 5

1.1 PURPOSE AND RESEARCH QUESTION 6
1.2 LIMITATIONS 7
1.3 DEFINITIONS 7
1.4 DISPOSITION 8

2. LITERATURE REVIEW 9

2.1 MISINFORMATION, DISINFORMATION AND FAKE NEWS 9
2.2 SOCIAL MEDIA AND DIGITAL VIGILANTISM 10
2.3 MULTIMODAL TOOLS 12
2.4 DESIGN AS A SOLUTION 13

3. METOD 15

3.1 LITERATURE SEARCH 15
3.2 CONCEPT DRIVEN DESIGN RESEARCH 16
3.2.1 SEVEN ACTIVITIES IN CONCEPT DRIVEN INTERACTION DESIGN 16
3.2.2 METHOD CHOICE 17
3.3 APPLICATION OF CONCEPT DRIVEN DESIGN RESEARCH 18
3.3.1 PARTICIPANTS 18
3.3.2 MATERIAL 18
3.3.3 PROCEDURES 19
3.4 ETHICAL CONSIDERATIONS 22

4. RESULTS AND ANALYSIS 23

4.1 SPECIFIC OPINIONS ABOUT DIFFERENT ASPECTS OF THE DESIGN 23
4.1.1 THE MAP 23
4.1.2 THE TIMELINES 23
4.1.3 NAVIGATION OF CAMERA ANGLES 24
4.1.4 COMBINING AUDIO WITH OTHER ANGLES 24
4.2 GENERAL OPINIONS OF THE CONCEPT 25
4.2.1 THE CONCEPT AS AN ENTIRETY 25
4.2.2 ETHICAL ASPECTS 27
4.2.3 OTHER AREAS OF USE 27
4.3 CONCEPT REVISITED 28
4.4 CONCEPT CONTEXTUALIZATION 29

5. DISCUSSION 30

5.1 RESULT DISCUSSION 30
5.2 METHODOLOGICAL DISCUSSION 32
5.3 ETHICAL IMPLICATIONS 32

6. CONCLUSION AND FUTURE RESEARCH 34

REFERENCES 36

APPENDIX 1 40

1. INTRODUCTION

In an age where people have access to mobile technology and its efficiency for diffusion, the public tends to take to social media for crisis communication (Jones, Garfin, Holman, & Silver, 2016; Simon, Goldberg, & Adini, 2015; Zubiaga, Liakata, Procter, Wong Sak Hoi, & Tolmie, 2016). Social media enables the access of real-time information of events to the public (Simon et al., 2015). The danger of this is the diffusion of fake news that often happens during times of crisis when people are the most vulnerable (K. Hunt, Agarwal, Al Aziz, & Zhuang, 2020; K. Hunt, Agarwal, & Zhuang, 2019), this especially since the debunking of said misinformation can be challenging and problematic (Ecker, Lewandowsky, & Apai, 2011). An example of an event where the diffusion of misinformation during an event surrounding a socially undesirable act became a safety hazard was during the attack at Utøya, Norway in 2011 (Frey, 2018).

The attack at Utøya was preceded by a bomb placed at the center of a Norwegian government office (Kolås, 2017). As Bernt Aardahl states in (Kolås, 2017) people tend to band together as an emotional response to events surrounding socially undesirable acts. Kolås (2017) raised the question of people's perception of national memory of a trauma, and how to preserve a just memory. Kolås (2017) states that social response to the trauma of the Utøya attack was undealt with and affected the *national text* over time, due to a continuous risk of misinformation.

Another event where fake news took place both during and after the event was at the Boston Marathon bombings in 2013 (Jones et al., 2016; Starbird, Maddock, Orand, Achterman, & Mason, 2014; B. Wang & Zhuang, 2018). Fake news that spread during this event included misidentifications of the bombers (Starbird et al., 2014) and the misidentification of an 8-year-old girl killed in the bombings (Jones et al., 2016; Starbird et al., 2014; B. Wang & Zhuang, 2018). The misidentification of these persons have been linked to digital vigilantism on social media (Myles, Benoit-Barné, & Millerand, 2018; Yardley et al., 2016). Trottier (2016) states that digital vigilantism is a process where citizens are collectively offended by other citizen activity, and coordinate retaliation on mobile devices and social platforms. According to Yardley et al. (2016) this practice engages in detective work including searching for information, uploading documents, images and videos, commenting, debating, theorising, analysing, identifying suspects and attempting to engage with law enforcement and other organisations and individuals connected to the cases. Both Yardley et al. (2016) and Chia (2018) states that the practice of digital vigilantism is growing and is becoming more and more

popular. When this practice fails in identifying the right persons of different events it results as problematic since digital vigilantism often includes “naming and shaming” of the targets (Trottier, 2016).

Previous studies have shown the advantage of combining both audio and video for segmentation and categorization of multimodal data such as video (Bouyahi & Ayed, 2020) during times of disaster (Pouyanfar, Tao, Tian, Chen, & Shyu, 2018). The technology behind the auditory analysis of said material is based on SoundNet, a deep convolutional architecture for natural sound recognition (Aytar, Vondrick, & Torralba, 2016; Bouyahi & Ayed, 2020; Pouyanfar, Tao, et al., 2018).

This study aims to propose a new interface that is based on the categorization of unlabeled data explored by Pouyanfar, Tao, et al. (2018) that combines the visual and auditory material of published videos that can display a more holistic view to the digital spectators of events surrounding socially undesirable acts. This interface is inspired by jigsaw puzzles and offers the ability of mapping auditory material from the scenario between the recorded visual material.

To summarize, the public will make conclusions of the information of events that they are witnessing digitally. Instead of exploring ways of stopping this from happening, this study intends to give the public access to more information. This means enabling the public to make more informed conclusions of events. By gathering information of this kind, users can evaluate and understand the context and characteristics from different physical aspects within the scene.

1.1 PURPOSE AND RESEARCH QUESTION

As digital vigilantism is a recurring phenomenon in events surrounding socially undesirable acts around the world (Chia, 2018; Yardley et al., 2016), there is a need for better tools to perform this practice. This need can be confirmed by recognizing that there have been various events where investigative efforts by digital vigilantists have resulted in misidentification of both perpetrators and victims (Chia, 2018; Trottier, 2016).

How can a design of a graphic interface for navigation of video material from social media theoretically be conceptualized to support digital vigilantists?

This research question will be answered through the method Concept Driven Design Research (Stolterman & Wiberg, 2010).

1.2 LIMITATIONS

This study will not investigate means of directly helping individuals involved in events surrounding socially undesirable acts with tools made for prognosis and evacuation but is focused on the public perception of said events. The study will not describe, nor form a case analysis of what has already been done in Pouyanfar, Tao, et al. (2018) that proves that basal media sourcing and combining deep learning techniques is applicable for the use and purpose of this study. However, the focus of this study lies in the exploration of an interactive interface that presents the video material that is retrieved through the concept of Pouyanfar, Tao, et al. (2018). The analysis for text descriptive materials will not be included as it is based on a separate data extraction process (H. Hao & Wang, 2020; Tian, Tao, Pouyanfar, Chen, & Shyu, 2018).

This study does not examine the effectiveness and accuracy that this application creates for digital vigilantists in their investigation but rather investigates the perception that digital vigilantists has for this type of application and how it should be designed with digital vigilantists as a target group.

1.3 DEFINITIONS

- *Socially undesirable act* - An act that causes the public to take offense with moral outrage (Chia, 2018).
- *Multimodal data* - (Pouyanfar, Yang, Chen, Shyu, & Iyengar, 2018) refers to multimodal data as textual, visual and acoustic data that represent different features from different media sources.
- *Fake news* - Lazer et al. (2018) defines Fake news as an expression that overlaps other information disorders such as misinformation and disinformation. This means that fake news is an expression for the spread of false information whether this is intentional or not.
- *Labeled data* - Labeled data is a designation for pieces of data that have been tagged with one or more labels identifying certain properties or characteristics, or classifications or contained objects. Labels make that data specifically useful in certain types of machine learning known as supervised machine learning setups (Techopedia, 2021).
- *Audio tagging* - Labeled audio data that describes audio events.

- *Deep learning* - Deep learning is a machine-learning algorithm that utilizes graph technologies with transformations among neurons to develop many-layered learning models (Pouyanfar et al., 2019).
- *Digital vigilantism* - Digital vigilantism is a process where citizens are collectively offended by other citizens´ activities and therefore respond through coordinated retaliation on digital media, including mobile devices and social media platforms (Trottier, 2016). This study therefore refers to digital vigilantism as a collective response to events surrounding socially undesirable acts where the public takes to digital media to find and expose the details of these events. This often includes identifying and exposing the perpetrators responsible for the socially undesirable acts.
- *Spatio-temporal data* – Usually involves the expression of spatio-temporal objects, its modeling, indexing and query of spatio-temporal data, the system structure, prototype systems and application of spatio-temporal database, and so on.,
- Ye, Peng, and Guo (2010) explains this as information about spatio-temporal objects. Spatio-temporal objects are objects whose boundaries or spatial position are changed over time. There are many practical needs of effective storing and management of spatio-temporal objects to query the objects' locations and their information in the past, at present, and to speculate their behavior in the future.

1.4 DISPOSITION

In the introduction, this paper presents a brief introduction on subjects of this paper. The main theme is fighting the spread of misinformation in the context of events surrounding socially undesirable events, and the tools that have previously been used for this purpose.

The literature review performs a deep dive into each subject and provides more detailed information of the background of earlier incidents and the technology that has been used to stop the spread of misinformation.

The method chapter consists of a description of the method *concept driven design research* that has been applied to realize the investigation of this paper.

After concept driven design reseaarch is explained the chapter **Results and analysis will present the findings** of the investigation. This is followed by a **Discussion** of the results from the earlier chapter and finally a **Conclusion** will sum up the accomplishments based on this study and propose further research fields for new investigations.

2. LITERATURE REVIEW

By gathering and evaluating information regarding the usage of social media in the context of fake news, this study proposes a new user interface as a tool for examining the validity of multimodal data online.

To examine fake news from a HCI (Human computer interaction) standpoint this study presents previous events of socially undesirable acts where social media acted as a crucial tool for communication and therefore also as a platform where fake news was spread and alternated the public understanding of the events (Pantumsinchai, 2018).

2.1 MISINFORMATION, DESINFORMATION AND FAKE NEWS

The spreading of false information is something that likely has been going on since humans started living in groups and has been transferred from mouth to mouth up until the printing press was invented (Burkhardt, 2017). The printing press gave new possibilities for spreading false information and an example of this is when the *New York Sun* published a series of articles of findings of life on the moon in 1835 (Allcott & Gentzkow, 2017). Since the 2016 election in the United States the word Fake News has gained mainstream attention (Allcott & Gentzkow, 2017). There are various different definitions and no set standard around the terminology of fake news, misinformation and disinformation (Domenico, Sit, Ishizaka, & Nunan, 2021). One version of these states that fake news is defined as fabricated information that mimics news media content in form but not in organizational process or intent (Lazer et al., 2018). Lazer et al. (2018) also describes it as an expression that overlaps misinformation (false or misleading information) and disinformation (false information that is purposely spread to deceive people). Previous studies have not been able to pinpoint a specific user group that is more apt to sharing and spreading des/misinformation (T. Buchanan, 2020; T Buchanan & Benson, 2019) but what has been found is that users have a tendency of sharing after having been exposed to the same information repeatedly (T. Buchanan, 2020). Another factor that affects users' trust in news is based on whether other users have reacted positively or negatively with the news (Colliander, 2019; Li & Sakamoto, 2014). When news, whether true or false creates a larger interaction by the public, they will gain more attention and become more visible and have thus generated what is called an organic reach (T. Buchanan, 2020; T Buchanan & Benson, 2019).

A problem with this is that false information often gains more attention than true information. According to Vosoughi, Roy, and Aral (2018) the amount of false information online is increasing and false news has 70 % more chance of being spread than true news. In their investigation the false news gained 1500 shares six times faster than real news on twitter. Harper et al. (2020) claims that the reason why false news gets more attention than real news is because false news often has a higher level of novelty than real news.

Rubin (2019) sees the problem of fake news as an epidemic and calls it the disinformation and misinformation triangle. This triangle is made of three key determinants. "Fakes" as Virulent Pathogens, "Online media" as Conducive Environments and "Gullible Readers" as Susceptible Hosts. Rubin (2019) thereafter suggests three interventions to disrupt the interaction of these factors.

1. **Automation** of identification of fakes
2. **Education** to inoculate the general public and create more informed citizens.
3. **Regulation** by discouraging social media platforms from pursuing short-term profit at the expense of human rights, democracy, scientific fact or public safety.

(Rubin, 2019)

The current study aims at taking on the misinformation problem by educating the public and creating more informed citizens.

2.2 SOCIAL MEDIA AND DIGITAL VIGILANTISM

Social media include a range of web applications such as social networking sites, picture- and video-sharing sites, weblogs, virtual worlds, and wikis. Social media have been studied in the aftermath of man-made as well as natural disasters and have been found to play a vital role in these events (Nilsen, Hafstad, Staksrud, & Dyb, 2018). In 2011 a single man dressed up as a police officer and went to a youth camp in Utøya, Norway and started shooting. The attack resulted in 69 deaths (Kolås, 2017) of the present 564 individuals on the island (Nilsen et al., 2018). Studies have shown that 94 % of contacted survivors shared their experiences in media (Thoresen, Jensen, & Dyb, 2014) and have listed 5 reasons for why survivors choose to use social media after the attack as: Information exchange; receiving and offering social support;

mourning; performing symbolic actions; and partaking in debates and discussions (Nilsen et al., 2018).

The rise of mobile technology and social media has allowed “ordinary” people from all over the world to become witnesses with events and interact with these by commenting and sharing online material (Wu & Montgomery, 2019). This results as problematic as social media users are notorious at rumor detecting during events such as disasters (Rubin, 2019; Starbird et al., 2014; Zubiaga et al., 2016) and have a tendency to rush to share early unverified information rather than waiting to share later verified information (Zubiaga et al., 2016). One of the key innovations of social media (over traditional media outlets) is that it provides users the ability to seek out information and share their own content within their own networks (Zeitzoff, 2017) which has allowed for the practice of digital vigilantism.

Digital vigilantism is a process where citizens are collectively offended by other citizen activity, and respond through coordinated retaliation on digital media, including mobile devices and social media platforms (Trottier, 2016). Other names of the same phenomenon include web sleuthing (Myles et al., 2018; Yardley et al., 2016), digilantism (Nhan, Huey, & Broll, 2015) and cyber vigilantism (Chia, 2018). The motives behind this practice have been studied by Yardley et al. (2016) who divide them into 11 categories with the three largest motives being to achieve justice or closure, uncover new leads or information, having been a victim of crime or affected by the type of case. Yardley et al. (2016) also divides this practice into what they call “solo web sleuthing" and “group web sleuthing” where the former refers to people doing their own private investigations of personal matters and the latter refers to a public and crowdsourced type of digital vigilantism.

The first well known event where a larger scale of digital vigilantism was used leads back to 2006 in china where internet users were able to identify a woman who in a video was crushing the skull of a kitten with her stiletto heels (Chia, 2018; Nhan et al., 2015). Although digital vigilantism has succeeded in identifying and exposing the perpetrators in events such as this, there have also been numerous events where the crowdsourced investigations have failed in their efforts of identifying the correct people involved. Examples of events where this has happened are the Boston bombing in 2013, the Bangkok bombing in 2015 and the Westminster attack in 2017. All of these are examples of when digital vigilantism on social media spread misinformation and false accusations (Myles et al., 2018; Pantumsinchai, 2018).

On April 15, 2013 two bombs exploded during the Boston marathon and resulted in 3 people being killed and another 264 being injured (Pantumsinchai, 2018). This event was followed by a digital manhunt of the perpetrators of the bombing. Digital vigilantists on social media wrongly identified two men as the perpetrators (Starbird et al., 2014). Four days after the bombings, FBI identified the suspects as the brothers Tamerlan and Dzhokhar Tsarnaev (Nhan et al., 2015; Starbird et al., 2014). The body of one of the wrongly accused suspects was found on April 23. His death had occurred weeks before the time of the bombings (Starbird et al., 2014).

A similar event occurred in London in 2017, when a man drove a car into a crowd of people killing 3 people and injuring over 50 persons. Digital vigilantists identified the perpetrator as a man who during the time of the attack was in prison (Pantumsinchai, 2018).

2.3 MULTIMODAL TOOLS

The fact that digital vigilantism has failed in completing it´s task of exposing correct information of events on social media therefore calls for a new solution of consuming the data that is uploaded by internet users of events surounding socially undesirable acts. Early access to fine-grained information helps to avoid uncertainties in events where there is a lack of authoritative information (H. Hao & Wang, 2020).

Previous research has explored the possibilities of retrieving datasets of multimodal data from social media in connection to disaster events (Pouyanfar, Tao, et al., 2018; Tian et al., 2018) and spatio-temporal analysis from surveillance cameras (Y. Hao, Xu, Liu, Wang, & Fan, 2018; Sreenu & Saleem Durai, 2019; J. Wang & Xu, 2016). This research has been mainly focused on optimizing the precision of material presented to users by using a wide range of deep learning techniques for analyzing the multimodal data (Sreenu & Saleem Durai, 2019).

H. Hao and Wang (2020) proposes a method for countering noisy, unstructured, and unbalanced social media data that eases the access of finding specific information that only resides in a few social media posts for events similar to disaster events. This creates the possibility of filtering and finding the adequate material for each specific event from social media platforms.

J. Wang and Xu (2016) and (Y. Hao et al., 2018) propose a framework for manifesting single angle CCTV (Surveillance camera system) data from video streams as a 3D image to predict, and detect anomaly behavior. This is achieved by gathering spatial visual data and slice samples

from crowded places at different spatial volumes, *spatio-temporal volumes* and further sampled at different volume levels to enrich the database for constructing the 3D image. Other previous works that include analyzing CCTV data is Machaca Arceda, Fernández Fabián, Laguna Laura, Rivera Tito, and Gutiérrez Cáceres (2016) that proposes a system for fast face detection in violent video scenes implying systems that has the possibility of automatically recognizing faces portrayed in videos from these video surveillance systems.

According to Tian et al. (2018) the detection and retrieval performance can be significantly improved by using various data types and that a human brain can detect events or concepts from a video by not only visualizing the spatio-temporal data but also listening to the audio. This being said, research has explored the possibilities, using sound as a navigating tool in a digital environment and the physical world, and even though audio tagging has not been sufficient for describing environments and associating to nearby places (Ibrahim, Royo-Letelier, Epure, Peeters, & Richard, 2020) sound tagging systems have shown great promise within an isolated music production context where it's possible to search and find sound in the same complex material with the same event characteristic (Fonseca et al., 2018). Pouyanfar, Tao, et al. (2018) investigates how combining audio and video tagging affects the duration findability of crucial information in regard to a 'disaster event'. They divide the information in; before, during and after the event. Extended measurements of time units have been done by dividing the content into early and late information etc. and concluded that the results of adding the factor of audio did not result as a significant improvement to the more effective tagging system that is the one of video. However, the system that Pouyanfar, Tao, et al. (2018) bases their sound model on SoundNet (Aytar et al., 2016), shows that this system is a powerful paradigm for learning sound representations from unlabeled videos.

Pouyanfar, Yang, et al. (2018) expresses the need for creating intuitive and interactive interfaces for viewing and interacting with multimodal database management systems and that interaction visualization tools are crucial to multimedia big data.

2.4 DESIGN AS A SOLUTION

The aim of this study is creating an interactive interface that is partly based on the concept of spatio-temporal data from different camera angles and extraction and combination of audio from video clips.

Schoeffmann, Hudelist, and Huber (2015) states that earlier research in the field of video interaction has mainly been focused on the user and on interface design aspects while on the other hand the research within the field of media retrieval has focused on content analysis and automatic search performance. Schoeffmann et al. (2015) expresses a need for future research to converge within these two fields to create more practical and usable video search tools.

Wijnants, Rovelo, Quax, and Lamotte (2016) proposes an application called *WanderCouch*, that creates an immersive experience of music festivals by combining professional and user generated content (UGC). To allow viewers to get a more immersive experience they created a graphic interface including the pages of: UGC playback screen, topographic map, panoramic map, and full screen concert playback. The UGC playback screen includes a browser that enables viewers to explore content published by users in the form of displaying thumbnails. The topographic map offers the possibility of seeing the entire festival area from above to get a mapped image of the space and to be able to move around in the space by clicking on different objects on the map. The panoramic view is explained as a "virtual navigation experience akin to Google Street View" that lets the viewer navigate through the festival area through UGC. Finally, the full screen concert playback shows the concerts as well as allows viewers to navigate through different angles through a map that shows from where the different angles are recorded and smaller windows of what those angles are showing in the moment.

The results of Wijnants et al. (2016) shows that even though *WanderCouch* does not replace the physical presence of a festival, viewers appreciated the experience of being able to visit a festival from home. It was expressed that UGC, and the panoramic map gave viewers a good understanding of the general theme and atmosphere while the topographic map gives an enhanced understanding of the spatial area of the festival as well as facilitating exploration. The option of viewing concerts from different viewpoints was appreciated and inspired viewers to deeply engage and interact with musical performances. Negative comments on the application included the quality of audio and video in UGC and that it was often meaningless content provided.

The appreciation of the possibility to view events from different perspectives has also been confirmed in Dezuli, Huber, Churchill, and Mühlhäusern (2013) and de Sá, Shamma, and Churchill (2013) who have both created platforms for video interaction by viewing events from perspectives through UGC. The idea of using maps as a way of orienting the user in the spatial surrounding environment has also earlier been implemented by Neng and Chambel (2010) who

uses a mini map to orient the user of its position and direction in a 360º hypervideo environment.

Karrer, Wittenhagen, and Borchers (2009) and Nguyen, Niu, and Liu (2014) have both explored direct manipulation video navigation. By this they mean the possibility to interact directly with objects within a video. An example is moving an object such as a football in a video to an earlier state and therefore rewinding the video to the moment that the football was located in said position (Karrer et al., 2009). The possibility of being able to interact in this direct way with a video opens up for a world of new possibilities for the consumption of video.

3. METHOD

3.1 LITERATURE SEARCH

In this study a literature search of scientific journal articles has been implemented. Rienecker and Stray Jørgensen (2017) explains that during literature search one should focus on three questions.

1. What do I need to know to be able to solve my assignment?
2. Where can I find the material that I need to solve my assignment?
3. How do I search for information for my assignment?

The articles that have been included in this study were selected by a process of three steps. First, they were chosen and added to thematic lists by their titles. Second their relevance was evaluated after a superficial skimming of the article. The final incorporation to this study was lastly decided after reading each article thoroughly.

The search has been performed via the databases HKR Summon and ACM Digital Library. The terms that have been used during this search includes “Fake news”,” misinformation”, “social media”, “digital vigilantism”, “digilantism”, “web sleuths”, “crowdsourcing”, “multimodal retrieval systems”, “deep learning”, “spatio-temporal data”,” video interaction” and different variations and combinations of these terms.

Preference has been given to articles that have been published within the last 5 years, but in cases where older articles have had a relevance they have also been included. In these cases, an aspiration has been held to sustain these articles with newer ones.

The HKR database allows for the searches to be filtered as peer-reviewed and therefore this filter has been applied to the searches via this database.

3.2 CONCEPT DRIVEN DESIGN RESEARCH

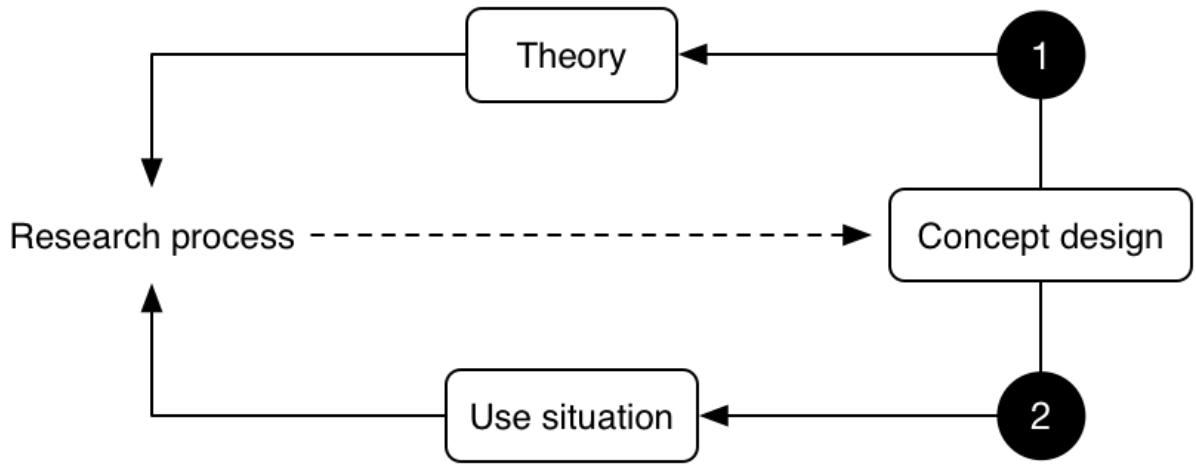


Figure 1. The flow of a concept driven design process. fetched from Stolterman & Wiberg (2010).

Based on the Stolterman and Wiberg (2010) method, Concept Driven design, this study examines the theoretical contribution to interaction design and user centered design. Given a theoretical tool to visualize and root the theory, validatiated concepts will build a ground for research by design concept representation. The design itself may contain sketch ideas or prototypes (Beaudouin-Lafon & Mackay, 2009; Lim, Stolterman, & Tenenberg, 2008) and the concept components have their own iteration through Concept driven design research.

Concept driven design research bridges as a tool and model for how users solve a design problem. This extends the discussion regarding interaction design and previous research. The components of the research process respond to users' needs and problems, manifesting as a design solution and as design research results (Stolterman & Wiberg, 2010).

The research design contribution connects to a concept or theory rather than an empirical study. A possibility to expand upon a hands-on approach and the research result does call back to a specific idea concept or standpoint in theory. The result points to a conceptual theory rather than specific user cases or user personas (Stolterman & Wiberg, 2010).

3.2.1 SEVEN ACTIVITIES IN CONCEPT DRIVEN INTERACTION DESIGN

The activities of the theory are; Concept Generation, Concept Exploration, Internal Concept Critique, Design of Artifacts, External Design Critique, Concept Revisited, Concept

Contextualization, and the cycle this study builds upon to present a specific result (Stolterman & Wiberg, 2010).

The flow of a concept driven design research process illustrated in **Figure 1.** is a proven method to connect the research question, theory and user context, that iterates the process of the seven activities (Stolterman & Wiberg, 2010).

Activity one, *Concept Generation*, idea iteration, background to mirror the research question, ideas, and sketches to further expand research. Activity two, *Concept Exploration*, evaluate material and the best iteration from step one, and form a more detailed context, to the concept. Activity three, *Internal Concept Critique* connects the concept to standpoints and the theoretical framework-specific example to conclude previous knowledge, by visualizing a prototype. Activity four, *Design of Artifacts* manifests the artifacts and their purpose for the theory and framework.

Activity five involves the test phase of *External Design Critique*, where potential, relevance and interaction take form with potential users. The concept should take in consideration the user's needs and theoretical standpoints. The design should manifest the uniqueness and extended problem solving that the concept gives the users. Activity six, *Concept Revisited*. The critique from the users should be further manifested in an updated version of the design and its concept, that meets the user needs with more precision and articulation. In activity seven, *Concept Contextualization* the concept manifests in its new form. The concept and its design contribution should be presented in comparison to other concepts and their design contribution, purpose and user context. The connection to theory, context and research should be manifested within the design concept. The design contribution should also further point to possible research and extended design contribution.

Stolterman and Wiberg (2010) concludes that good design manifests uniqueness for its context and design contribution. Good design builds upon a theory of research questions and previous techniques for the context of the concept.

3.2.2 METHOD CHOICE

A benefit of concept driven design research is the ability to theorize the use of a design concept without having the need of the technology for the practical manifestation (Stolterman & Wiberg, 2010). In this way one can manifest a design concept theoretically (Lim et al., 2008). By approaching the purpose of this study from a design perspective, concept driven design

research allows for designers to imagine and manifest new possible futures (Lim et al., 2008; Stolterman & Wiberg, 2010) within the space of a scientific research (Stolterman & Wiberg, 2010).

The literature search in this study therefore served as a base of information about the previous research of other similar artefacts and of the insights that this has resulted in. Therefore, this earlier research has guided the initial design that has been proposed for the execution of this study.

3.3 APPLICATION OF CONCEPT DRIVEN DESIGN RESEARCH

3.3.1 PARTICIPANTS

Five participants took part in the evaluation of the design concept in this study. All participants have had different experiences with performing digital vigilantism and are therefore included in the concepts target audience. The participants were exclusively male and between the ages 27-40 years old.

For the acquisition of participants, these were contacted via different Facebook groups where digital vigilantism takes place.

3.3.2 MATERIAL

The evaluation of the design concept was done by holding semi structured interviews. Semi structured interviews are a qualitative method where the questions give space for elaborated answers and do not necessarily need to be asked in a specific order (Patel & Davidson, 2019). This allows for participants to openly discuss the design aspects of the design concept on a higher level of abstraction, as done in the evaluation of *WanderCouch* by Wijnants et al. (2016).

The interviews were all held in video calls via Zoom or Facebook Messenger. A Figma prototype based on a scenario was sent to the participants who explored the prototype while sharing their screen and having the possibility to ask potential questions to the interviewer. After having tried the prototype to fully understand the concept the interviewer began the interview. The participants were allowed to review the prototype while answering the questions. The interview consisted of 12 questions (Appendix 1) that are divided into two categories: more *specific opinions about different aspects of the design* as well as *general opinions of the concept.*

During the acquisition of participants and in the beginning of each meeting with participants they were informed of the ethical considerations (see chapter 3.4) of this study. All participants agreed to participate under these conditions.

3.3.3 PROCEDURES

The first activity of concept driven is *Concept Generation.* This activity was held at an abstract level and was driven by encountering what fields of research that would be relevant to this specific case by exploring earlier research. Stolterman and Wiberg (2010) states that it is valuable that this activity is based on earlier research and that a distinct quality is to search for the unexpected.

As the second activity, *Concept Exploration,* calls for an exploration of materials and finding new design spaces (Stolterman & Wiberg, 2010), different tools for manifestation of the concept were explored and discussed. The possibilities that these tools allow for generated new ideas for the design and manifestation of the concept. Tools explored at this stage were mainly Eco studio, Unity and Figma. The result of this activity was four primitive prototypes that manifested a scenario and the initial ideas of what should be designed.

After the creation of these first prototypes, the *Internal Concept Critique* activity began. This implied identifying the uniqueness of the chosen core concepts and to what extent the concepts relate to existing theory, and how well these concepts can be clearly expressed in a concrete design (Stolterman & Wiberg, 2010). This resulted in an identification of further need for previous research in the design area of video interaction tools.

With a broader understanding of this previous research, the activity of *Design of Artefacts* could be engaged. The goal of this activity was to create a concrete manifestation that incorporates the concept design as a" whole" (Stolterman & Wiberg, 2010). This led to the prototype that was used for the user study of this paper. The prototype was made in Figma and represented an interactive video player where the user has the possibility to interact directly with objects in the video that lets them change camera view or to choose to only hear the audio from that other camera (**Figure 2.**).

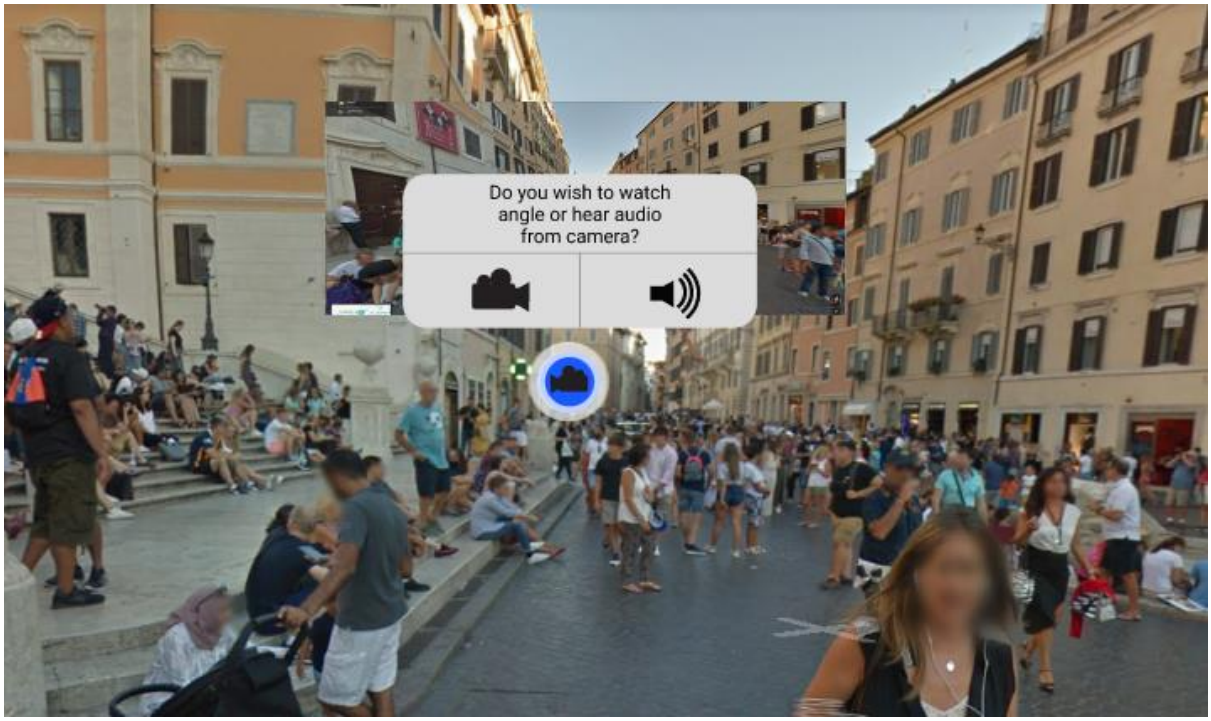


Figure 2. Video or sound from another angle.

The prototype also includes a "live" map that shows what cameras are available in the area and the temporal direction of each camera as well as the possibility to interact with this camera (**Figure 3.**).



Figure 3. The map, available cameras and audio.

Thirdly it also included both an individual timeline as well as a general timeline that portrays the time aspect of each individual video compared to the other videos available (**Figure 4.**).

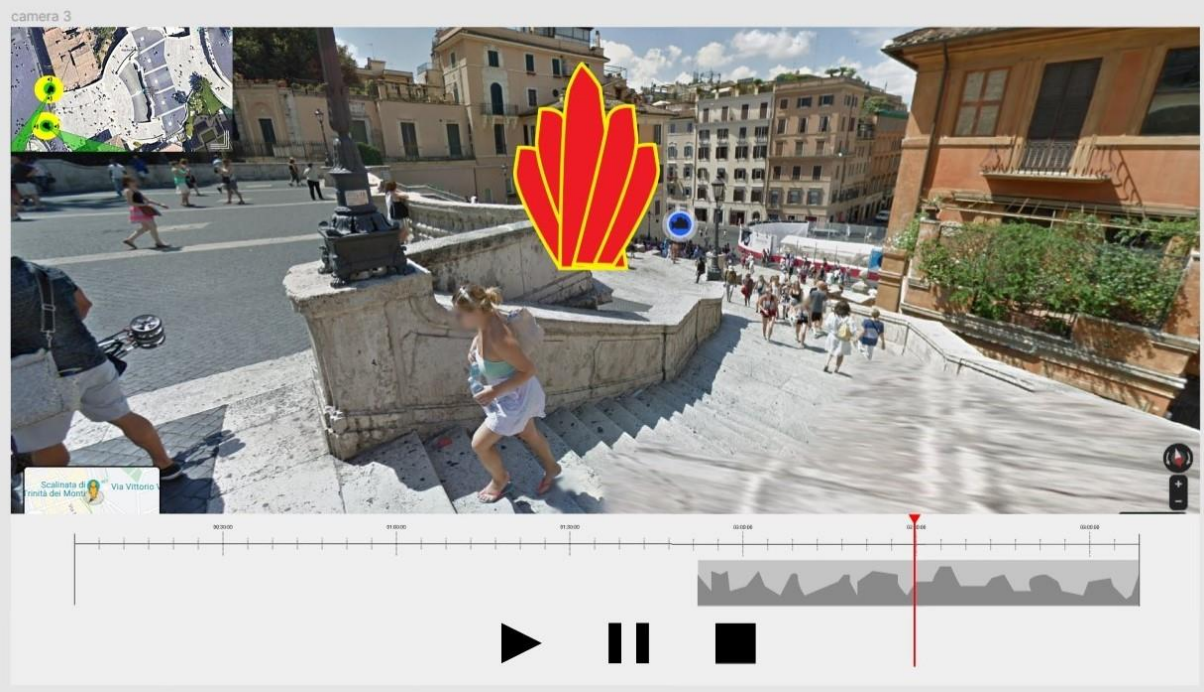


Figure 4. Individual videos and timelines.

The activity of *External Design Critique* refers to testing and was done through the user study. Stolterman and Wiberg (2010) expresses that testing means exposing the conceptual design to a public and having it critiqued as a composition. This was performed by letting users try the prototype and asking them questions of how they experienced it.

In *Concept revisited* the concept is revised after the insights that the critique has resulted in (Stolterman & Wiberg, 2010). These insights are all presented in chapter **4.3** (Concept Revisited).

Stolterman and Wiberg (2010) argues that after the final concept has been defined the final activity of *Concept contextualization* begins. This involves comparing the concept to previous concepts and theory and positioning the concept among these. Similar concepts involve WanderCouch by Wijnants et al. (2016) that incorporates a similar video player but in a different use case. Other concepts that involve similar technology are CCTV (surveillance camera) systems. These however do not include the factor of mobile cameras and social media

for content generation. The result of activity is further described in chapter **4.4** (Concept Contextualization).

3.4 ETHICAL CONSIDERATIONS

In 2017, The Swedish Research Council published 4 concepts of ethics within research. *Secrecy, Professional secrecy, Anonymity* and *Confidentiality*.

The concept of *Secrecy* states that "*The basic principle is that public documents shall be publicly accessible and that information can be covered by secrecy only if falls under a specific provision of the Public Access to Information and Secrecy Act.*"

(Swedish Research Council, 2017)

The concept of *Professional secrecy* states that "*Professional secrecy follows from secrecy to the extent that if certain information is covered by secrecy, then this also entails a requirement of professional secrecy about it*."

(Swedish Research Council, 2017)

The concept of *Anonymity* states that "*In some cases the anonymizing of information is a condition set by an ethics review board for its approval of a study. This can be done, for example, by removing personal information on completed questionnaires or samples.*"

(Swedish Research Council, 2017)

The concept of *Confidentiality* states "*The importance of confidentiality and of the researcher taking measures to protect the integrity of research subjects and their right to protection of their private lives.* "

(Swedish Research Council, 2017)

In this study, participants were informed that their identity would be kept anonymous, but that their age, gender and answers during the interview would be published in the study.

4. RESULTS AND ANALYSIS

In the following chapter the results and answers of the participants will be presented as well as an analysis of what these results imply.

4.1 SPECIFIC OPINIONS ABOUT DIFFERENT ASPECTS OF THE DESIGN

4.1.1 THE MAP

While all users agreed to the functionality of the map. Three of the participants did not have any negative comments about it while two of the participants found the presentation of the map disturbing.

"The big map feels too much, I would prefer to be able to hide it. I would also like to be able to zoom and have different textualizations of the map, like 3D satellite etc."

- Participant 5

"It feels like the map is in the way, maybe you could hide it."

- Participant 4

The other participants expressed a positive attitude towards the map in their comments.

"I couldn't imagine how it would work without the map. It's essential, even though the photo and video is a major improvement, knowing of the position and direction is amazing."

- Participant 2

4.1.2 THE TIMELINES

All participants expressed a positive attitude towards the timelines. However, participants 1 and 2 both expressed new suggestions for improvement upon this element.

"It's Nice to have that control second by second. Controlling from different angles. It would be cool to see all angles at the same time in a window so I can see the timeline going on at the same time, that would give you more control."

- Participant 1

“It's necessary, it helps a lot to situate the event and to be able to use all perspectives. It would be better if it used global time.”

- Participant 2

Other suggestions on the timeline arose during the interviews that included making the timeline more dynamic and implementing a function where users can leave markers to keep track of important events in the videos.

4.1.3 NAVIGATION OF CAMERA ANGLES

All participants stated that this is what is the concept's main feature.

“It's really nice. I have never seen anything like it. That makes it interesting. I mean it's kind of the point of the application… isn´t it?”

- Participant 3

However, this aspect of the concept also received suggestions as:

“As I said I think that this is the most important feature, a possible development could be to identify people that are there that you know. To find out if they are nearby. “

- Participant 4

“Zooming. I'd like to zoom in. and maybe select certain objects and be able to zoom in and out of them.”

- Participant 1

4.1.4 COMBINING AUDIO WITH OTHER ANGLES

The function of being able to combine audio with video received very different reactions from participants. Two of the participants did not see the need for the funcion while the other three participants expressed a very positive attitude towards it.

“There is no use of this, it's too real”

- Participant 4

“This is really useful, especially if you can use the camera from one angle and the sound from another. Then you can have the possibility to mix and match from different angles. Even if the sound is of bad quality or if the camera and sound have different distances.”

- Participant 5

Participant 1 had a very positive attitude towards the audio feature but also gave some suggestions for improvement.

“I would want to know where the sound is coming from and more control for toggling the sound on and off from different camera angles. I don’t know why you would listen to all angles at the same time, but it would be nice to know that I could.”

- Participant 1

4.2 GENERAL OPINIONS OF THE CONCEPT

4.2.1 THE CONCEPT AS AN ENTIRETY

All participants shared a positive view and excitement for the entirety of the concept. Concerns that arose included:

“I think it's too good. I see the danger and the ethical problems that arise with it. Personally, I would love to use it. Maybe not always for the correct reasons but I think it's really cool.”

- Participant 2

“I think that it is still in a very early stage. You still have a long way to go and different aspects that you need to work out before actually creating it. But i love the idea. I think that this could be a really useful tool in many different ways.”

- Participant 5

Other thoughts on the concept as a whole included different ideas of what users would be able to do with the technology and for whom it might be interesting.

"I think that it could be very interesting for curious people, investigators and historians. In general, for people who prefer graphic information. It is a very good way to solve it. Facebook for example bombs you with videos that you really have no idea where they come from. You don't know what happened. Is this meant to be a social network?"

- Participant 3

"This could be used by agencies. Measure things like blast radius. Discover the type of bomb. Details of actual explosion. Use for future surveillance of the area."

- Participant 1

When asked if anything were missing in the concept three of the five participants expressed that they would like to have the possibility to hide the graphic elements in the video so they could see the footage without having distractions.

"I think it would be beneficial to be able to hide all functions like camera icons and the map and everything, if I'm watching the video, I don't want to miss anything that happens that could be crucial information"

- Participant 5

Other suggestions that came up include making a space where users can publish animations of angles that were not filmed and similar material, also including the possibility to interact and share tips with other users was suggested. One participant saw a need of setting up security protocols, such as user identification and restricting the area of events until users request for the area to expand.

When asked if they experienced anything as redundant in the concept none of the users expressed that they experienced anything as redundant.

When asked of how their opinion of consuming video material in this concept in comparison to on other social media platforms no negative comments arose. Four of the users focused their answers on how this concept offers the users an enhanced control.

"I´d prefer to use this application. It lets me see all the angles, pause, and combine. Maybe I could take photos as memories and zoom in. I think this application has the potential to use all the stuff that is missing on pages like YouTube."

- Participant 3

"The strength in the concept is that it is unique and with the development of camera angles and the live perspective in the map makes it stand even stronger"

- Participant 5

4.2.2 ETHICAL ASPECTS

The general answer from all participants about the ethical implications of the concept expressed that there will always be people who will try to use technology with malicious intentions. Three of the participants expressed concerns about the privacy of the people portrayed in the video footage.

Only one of the participants more elaborated concerns of the ethical implications.

"How would you choose what events to portray and what events not. Maybe the users can request for events to be opened up and when you reach a minimum number, this shows this event. I do see a danger in it for stalkers as you can follow people's movements. Especially if this technology gets in the wrong hands. The problem here could come out of employees being corrupt or if somebody hacks the application. You would have to make sure you have watertight security. Maybe users should have to identify themselves or something,"

- Participant 2

4.2.3 OTHER AREAS OF USE

Three of the participants named sports events and concert events as possible areas of use.

“Massive events. I think this would work for concerts where people want to watch a certain video with better sound quality that might come from another camera. Sport events is another one.”

- Participant 3

Two of the users expressed that this concept could be used by military, police and homeland security.

” For the government. It is a good tool for the military, police, and homeland security. Basically, it's a way to make security cameras better. “

- Participant 2

Other ideas of potential areas of use that arose during the interviews included Esports and the exploration of historical sites.

4.3 CONCEPT REVISITED

The results presented above bring new insights of changes that should be implemented in the concept of this study, Figure Regarding the general idea of the concept there was a general acceptance of the concept as a whole.

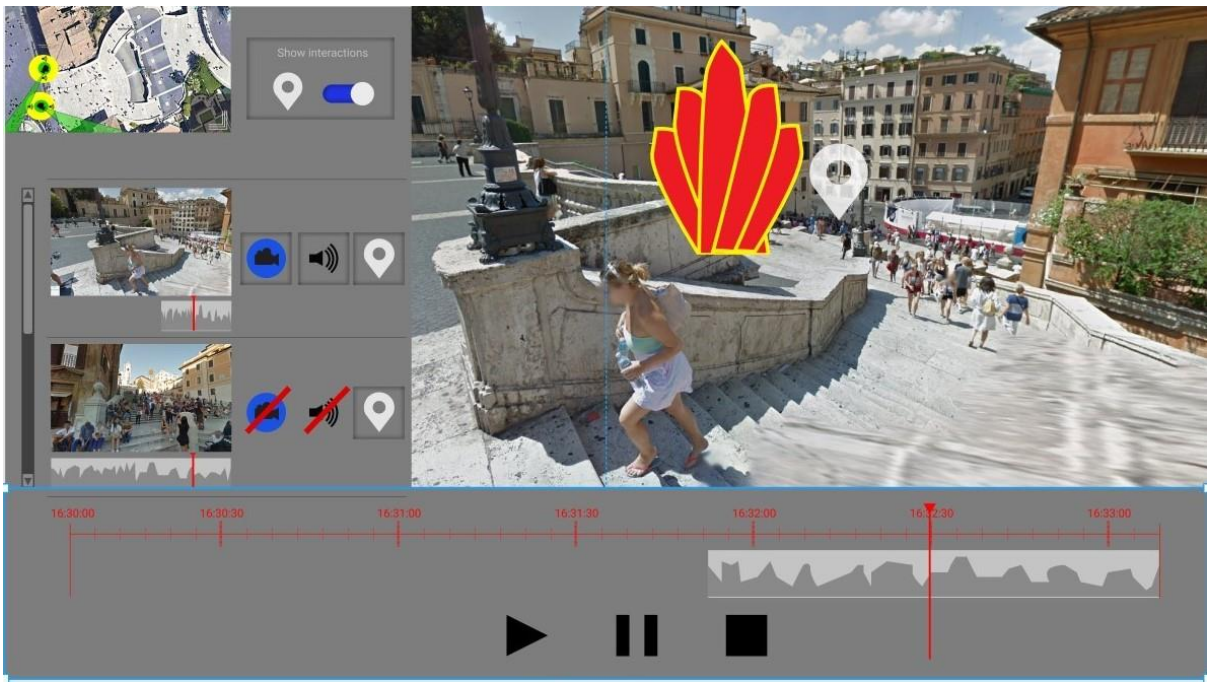


Figure 5. Updated concept

However, a major part of the participants expressed a need for being able to hide items in the graphic interface of the video since they disturbed and distracted users from seeing what was going on in the scene. An element that received criticism by two of the participants was the map with the arguments that it was too big and occupied too much space in the video. The major proposal for change in the graphic interface of the concept therefore involves creating a space outside of the video screen for placing the map and toggling buttons for hiding and showing the interaction points for objects in the video.

The creation of this space also allows for implementing other suggestions that were proposed by individual participators. Such as, showing windows of other angles in miniature windows, extended controls, and indicators from where the sound is playing.

The time of the timeline will be changed to the global time to indicate at what exact time the different events took place. As the windows of the other angles will be placed to the side of the video, they will also display their individual timeline to inform the users of the orientation of events in each video.

Another point that was requested by four of the participants was the ability to zoom and therefore this will also be added to the updated version of the concept on both the map and in the video elements.

The possibility of creating a social network around this concept would imply a larger change of direction and would need further research into the subject to develop. However, the possibility of this change will not be excluded. The ethical aspects of the design that was brought forward during this study included mainly privacy and security. This is an area that will also need further research, to be able to make a preliminary proposal of a solution to these problems.

4.4 CONCEPT CONTEXTUALIZATION

Wijnants et al. (2016) proposed a prototype in their article that is called *WanderCouch*. This prototype holds a high level of similarity in the graphic interface as the concept proposed in this study is inspired and has been designed from many of the insights that were discovered in Wijnants et al. (2016). The largest difference of these two prototypes lies within the context that they are used. While *WanderCouch* is focused on experiencing festivals and exploring its

areas, the concept of this study is meant for finding out what happened and exploring scenes of events surrounding socially undesirable acts.

Pouyanfar, Tao, et al. (2018) Proposes their new multimodal deep learning framework for video classification. This framework collects multimodal data and classifies the data that allows for a retrieval of data that comes from a specific time and place. The data is later revised to evaluate the developments of events. This framework is a building block for the concept that is presented in this study. The way that this concept elaborates on top of the Pouyanfar, Tao, et al. (2018) framework is by creating a graphic interface that allows for users to navigate through the available data.

J. Wang and Xu (2016) and Y. Hao et al. (2018) developed frameworks for creating 3D images out of a single CCTV angle to detect anomaly behavior. Other solutions for using surveillance cameras are home security camera systems such as Arlo (Arlo, 2021). However, these systems and frameworks do not involve mobile camera videos that are published on social media and are therefore restricted to use the angles statically placed cameras which this concept do not.

5. DISCUSSION

5.1 RESULT DISCUSSION

All participants in this study presented a positive attitude towards the concept in general. This confirms that the idea of the concept is well received and appreciated by the target group intended for the concept, and that this would be an application that they actually would use instead of consuming the same material via other social media platforms. However, participants also expressed that they believed that the concept is still at a very early stage, that there is still a long way to go and many considerations to take in mind before this concept actually can be realized.

One of the major insights of the concept that this study led to was that the icons and elements that were placed on top of the video to indicate that interaction is possible were experienced as disturbing by the majority of the users. This was a concern that was not expressed by Wijnants et al. (2016) that included both the map and interaction points to change camera angle nor by (Dezuli et al., 2013) that displayed icons on top of the screen during sports events. This could be because of the design implemented of the map and these icons in this study, but participants

clearly stated that they did not want to miss what was happening on the screen. Therefore, this leads to the conclusion that for the context of use for digital vigilantists, the screen showing the video needs to have the possibility of being cleared from other visual elements.

The map and its ability of showing where and what direction is being filmed was positively received and therefore concords with both Wijnants et al. (2016) and Dezuli et al. (2013) that both use this kind of maps in their prototypes.

The function of being able to navigate through different camera angles both directly through the video as well as via the map was seen as the main feature by all participants. This positive reaction confirms the functionality of the concept that is based on the prototypes presented in Wijnants et al. (2016) and Dezuli et al. (2013).

The possibility of combining sound from different camera angles was not received in unison by participants. This leaves it as a part of the concept that could still be kept for further evaluation with more possible users.

A proposal for a space where users can publish animations of not recorded angles, the detonation of the bomb and similar, confirms an idea that was disregarded as redundant at an earlier stage of the design of the concept. This involved a presentation of spatio-temporal data that has the ability to create 3D images of movements to detect crowd anomaly behaviour. The framework of this system is developed by J. Wang and Xu (2016) and Y. Hao et al. (2018). The creation of new spaces for presenting other information could be beneficial as the practice of digital vigilantism includes more work than only revising videos (Yardley et al., 2016).

Alternative areas of use that were revealed during this study were mainly sports events and musical events. This reflects and further confirms Wijnants et al. (2016) that approaches musical festivals with a similar concept and Dezuli et al. (2013) that approaches sports events with a similar concept. The other area of use that was proposed was to maintain the same concept, but to change the target group to military, homeland security and police. Implementing this change could possibly rid the concept of many ethical implications as the use would be restricted to a more controlled context. This would follow the statement by Trottier (2016) that the public should only support the police by providing data to the investigation but that only the police should perform the actual investigation.

Whether this application can help to battle the spread of fake news among digital vigilantists is a subject for further investigation. This study confirms that this would be a preferred tool for performing investigations. However, this does not necessarily imply that it actually is a more

effective tool. As this study intended to inoculate the public with the aim of creating more informed citizens as an intervention according to the disinformation and misinformation triangle (Rubin, 2019). This would by preference be investigated in a study with means of measuring the accuracy of the investigations held by digital vigilantists using this application in comparison to other digital media.

5.2 METHODOLOGICAL DISCUSSION

The method chosen for this study is Concept driven design research (Stolterman & Wiberg, 2010). This decision was based on the idea of creating a concept to fill an observed void of existing tools that allows digital vigilantists to navigate through content in an effective way. As designers, the authors of this study saw it fit and experienced the benefits of this method, since it allowed them to use their experience and knowledge within the area of design to perform this study. This method affords to theoretically and visually manifest both the processes and the final results of a design simultaneously as it takes root in earlier research.

For the implementation of the study, qualitative and semi structured interviews were implemented (Patel & Davidson, 2019). The benefits of this method were the possibility of getting elaborated answers on both the more specific questions as well as getting insights on the participants general thoughts of the concept. This led to insights on questions that were not specified in the pre-defined questions for the interviews.

The search of the participants resulted in an all-male selection with ages ranging from 27- 40 years old. This is not a broad selection of participants. However, these were the persons who responded to the calling for the study and could therefore arguably be fit to represent the target group of digital vigilantists.

5.3 ETHICAL IMPLICATIONS

As earlier expressed by (Pantumsinchai, 2018) and Myles et al. (2018) the practice of digital vigilantism has not always led to accurate identifications of perpetrators and victims. This implies that there is a need for further research into whether this is a technology that not only can be developed but also if it really should be developed. Trottier (2016) expresses this need by saying that it remains an area of concern whether social actors should have the possibility to appropriate police or not in this digital environment. Digital vigilantism has in certain cases

supported police in their investigations and has in other cases interrupted police work Trottier (2016). This could imply a new direction of research in investigating ways to constrain the public from interrupting police work and using these types of concepts in unethical ways. The way that the social media platform Reddit decided to solve this problem was to enforce rules that discouraged future initiatives of divulgation of personal information belonging to third parties, like suspects and punitive self-justice among its members (Myles et al., 2018). However, this is not the direction taken in this study. The enhancement of a video player which gives digital vigilantists more control of the published material could perhaps lead to more accurate guesses for both victims and perpetrators of specific events. Therefore, it might be a benefit in combining this concept with a rigorous set of rules of use, as done in the case of reddit (Myles et al., 2018). This would also go in the line of the third intervention (Regulation) of the disinformation and misinformation triangle, (Rubin, 2019) which involves discouraging social media platforms from pursuing short-term profit at the expense of human rights, democracy, scientific fact or public safety.

Another issue to take in mind would be the selection of what events should be portrayed and what area should be included in the portrayal of these events. In this study a suggestion was proposed to leave this up to the platform's users, however this is a proposal that would need to be investigated before it was to be incorporated in this concept. Another interesting approach would be to investigate the possible integration of the system ViEWS (Hegre et al., 2019) that has the ability of forecasting conflicts based on maps. Depending on the development of precision in this system, it could be a deciding factor of what conflicts and what areas within these conflicts should be portrayed.

Yet another aspect that should be considered is how this concept could affect digital vigilantists. According to Yardley et al. (2016) web sleuthing is something that can turn into an obsession and result in difficulties for the vigilante to hold apart their personal life from the lives of others that they are investigating.

6. CONCLUSION AND FUTURE RESEARCH

The purpose of this study is to support digital vigilantists in their navigation through video material so that they can come to more informed conclusions of events surrounding socially undesirable acts. A need for new tools has been identified through searching previous research that states that digital vigilantism is a growing practice that in many cases results in misidentifications of both victims and perpetrators (Yardley et al., 2016).

To answer the question "*How can a design of a graphic interface for navigation of video material from social media theoretically be conceptualized to support digital vigilantists?",* this study has proposed a prototype of how this can be conceptualized. This prototype manifests the insights that have been reached in this study through Concept driven design research (Stolterman & Wiberg, 2010).

This study is not answering the question whether this application improves the accuracy of the investigations performed by digital vigilantists, but rather how this application is received by digital vigilantists. As the goal of the application is to battle spread of fake news among digital vigilantists, this study recognizes a need for future research of the difference in effectiveness and accuracy that this or similar applications could bring to digital vigilantists or other investigators. For this purpose, this study offers a foundation of insights of how an application could be designed for further research into the subject.

This study confirms that the concept being proposed is attractive and that digital vigilantists would rather prefer to use this tool in their investigative engagements. However, this study also recognizes various possible ethical implications and therefore reaches the conclusion that there is still a need for extensive research into whether this concept should be realized, or if there are possible ways to constrain the users into an ethical use of the design. The ethical concerns of digital vigilantism are also shared in previous research (Myles et al., 2018; Trottier, 2016; Yardley et al., 2016). The need for continuous research within the ethical aspects includes the effects that the practice has on both the users and the people portrayed in the videos.

This study presents a range of other technologies that could possibly be further investigated for a possible implementation into this concept to improve upon the experience and the ability to perform digital vigilantism in a more precise way. These include the crowd anomaly detection systems created by J. Wang and Xu (2016) and (Y. Hao et al., 2018) and the face detection system in violent scenes that is presented by (Machaca Arceda et al., 2016).

Alternative fields where this technology could be used is another subject for future research. Participants within this study expressed specifically that they saw a certain use for this technology in areas such as musical events, sports events and for governmental use. Similar concepts within the fields of musical and sports events have already been explored by Wijnants et al. (2016) and (Dezuli et al., 2013). However, other research fields including nonviolent events could distance the concept from many of the ethical implications that arise when focusing on violent and sensitive events.

REFERENCES

Allcott, H., & Gentzkow, M. (2017). Social Media and Fake News in the 2016 Election. *Journal of Economic Perspectives, 31*(2), 211-236. doi:10.1257/jep.31.2.211

Arlo. (2021). Arlo. Retrieved from https://www.arlo.com/

Aytar, Y., Vondrick, C., & Torralba, A. (2016). SoundNet: learning sound representations from unlabeled video. *NIPS'16: Proceedings of the 30th International Conference on Neural Information Processing Systems*, 892–900. doi:10.5555/3157096.3157196

Beaudouin-Lafon, M., & Mackay, W. (2009). Prototyping Tools and Techniques. *Human-Computer Interaction*, 137-160. doi:10.1201/9781420088892.ch7

Bouyahi, M., & Ayed, Y. B. (2020). Video Scenes Segmentation Based on Multimodal Genre Prediction. *Procedia Computer Science, 176*, 10-21. doi:10.1016/j.procs.2020.08.002

Buchanan, T. (2020). Why do people spread false information online? The effects of message and viewer characteristics on self-reported likelihood of sharing social media disinformation. *PLoS One, 15*(10), e0239666. doi:10.1371/journal.pone.0239666

Buchanan, T., & Benson, V. (2019). Spreading Disinformation on Facebook: Do Trust in Message Source, Risk Propensity, or Personality Affect the Organic Reach of "Fake News"? *Social Media + Society, 5*(4). doi:10.1177/2056305119888654

Burkhardt, J. M. (2017). Chapter 1 history of fake news. *Library Technology Reports, 53*(8).

Chia, S. C. (2018). Crowd-sourcing justice: tracking a decade's news coverage of cyber vigilantism throughout the Greater China region. *Information, Communication & Society, 22*(14), 2045-2062. doi:10.1080/1369118x.2018.1476573

Colliander, J. (2019). "This is fake news": Investigating the role of conformity to other users' views when commenting on and spreading disinformation in social media. *Computers in Human Behavior, 97*, 202-215. doi:10.1016/j.chb.2019.03.032

Council, S. R. (2017). *Good Research Practice*. (978-91-7307-354-7). Stockholm: Vetenskapsförådet.

de Sá, M., Shamma, D. A., & Churchill, E. F. (2013). Live mobile collaboration for video production: design, guidelines, and requirements. *Personal and Ubiquitous Computing, 18*(3), 693-707. doi:10.1007/s00779-013-0700-0

Dezuli, N., Huber, J., Churchill, E. F., & Mühlhäusern, M. (2013). *CoStream: co-construction of shared experiences through mobile live video sharing. Proceedings of the 27th International BCS Human Computer Interaction Conference*. Paper presented at the BCS Learning & Development Ltd.

Domenico, G. D., Sit, J., Ishizaka, A., & Nunan, D. (2021). Fake news, social media and marketing: A systematic review. *Journal of Business Research, 124*, 329-341. doi:10.1016/j.jbusres.2020.11.037

Ecker, U. K., Lewandowsky, S., & Apai, J. (2011). Terrorists brought down the plane!--No, actually it was a technical fault: processing corrections of emotive information. *Q J Exp Psychol (Hove), 64*(2), 283-310. doi:10.1080/17470218.2010.497927

Fonseca, E., Plakal, M., Font, F., Ellis, D. P. W., Favory, X., Pons, J., & Serra, X. (2018). General-purpose tagging of freesound audio with audioset labels: task description, dataset, and baseline. *In Proceedings of the Detection and Classification of Acoustic Scenes and Events 2018 Workshop (DCASE2018)*, 69–73.

Frey, E. (2018). "Do You Tweet When Your Friends Are Getting Shot?" Victims' Experience With, and Perspectives on, the Use of Social Media During a Terror Attack. *Social Media + Society, 4*(1). doi:10.1177/2056305117750715

Hao, H., & Wang, Y. (2020). Leveraging multimodal social media data for rapid disaster damage assessment. *International Journal of Disaster Risk Reduction, 51*. doi:10.1016/j.ijdrr.2020.101760

Hao, Y., Xu, Z.-J., Liu, Y., Wang, J., & Fan, J.-L. (2018). Effective Crowd Anomaly Detection Through Spatio-temporal Texture Analysis. *International Journal of Automation and Computing, 16*(1), 27-39. doi:10.1007/s11633-018-1141-z

Harper, L., Herbst, K. W., Bagli, D., Kaefer, M., Beckers, G. M. A., Fossum, M., . . . committee, E. r. (2020). The battle between fake news and science. *J Pediatr Urol, 16*(1), 114-115. doi:10.1016/j.jpurol.2019.12.004

Hegre, H., Allansson, M., Basedau, M., Colaresi, M., Croicu, M., Fjelde, H., . . . Vestby, J. (2019). ViEWS: A political violence early-warning system. *Journal of Peace Research, 56*(2), 155-174. doi:10.1177/0022343319823860

Hunt, K., Agarwal, P., Al Aziz, R., & Zhuang, J. (2020). Fighting fake news during disasters. *OR/MS Today, 47*(1), 34–39.

Hunt, K., Agarwal, P., & Zhuang, J. (2019). Tracking storms of misinformation spread amid disasters. *ISE Magazine, 51*(9), 28-32.

Ibrahim, K. M., Royo-Letelier, J., Epure, E. V., Peeters, G., & Richard, G. (2020). AUDIO-BASED AUTO-TAGGING WITH CONTEXTUAL TAGS FOR MUSIC. *2020 IEEE International Conference on Acoustics, Speech and Signal Processing*. doi:10.5281/zenodo.3648287

Jones, N. M., Garfin, D. R., Holman, E. A., & Silver, R. C. (2016). Media Use and Exposure to Graphic Content in the Week Following the Boston Marathon Bombings. *Am J Community Psychol, 58*(1-2), 47-59. doi:10.1002/ajcp.12073

Karrer, T., Wittenhagen, M., & Borchers, J. (2009). PocketDRAGON: a direct manipulation video navigation interface for mobile devices. In Proceedings of the 11th International Conference on Human-Computer Interaction with Mobile Devices and Services. *Association for Computing Machinery, 47*, 1-3. doi:10.1145/1613858.1613917

Kolås, Å. (2017). How critical is the event? Multicultural Norway after 22 July 2011. *Social Identities, 23*(5), 518-532. doi:10.1080/13504630.2016.1271740

Lazer, D. M. J., Baum, M. A., Benkler, Y., Berinsky, A. J., Greenhill, K. M., Menczer, F., . . . Zittrain, J. L. (2018). The science of fake news. *Science, 359*(6380), 1094-1096. doi:10.1126/science.aao2998

Li, H., & Sakamoto, Y. (2014). Social impacts in social media: An examination of perceived truthfulness and sharing of information. *Computers in Human Behavior, 41*, 278-287. doi:10.1016/j.chb.2014.08.009

Lim, Y.-K., Stolterman, E., & Tenenberg, J. (2008). The anatomy of prototypes. *ACM Transactions on Computer-Human Interaction, 15*(2), 1-27. doi:10.1145/1375761.1375762

Machaca Arceda, V. E., Fernández Fabián, K. M., Laguna Laura, P. C., Rivera Tito, J. J., & Gutiérrez Cáceres, J. C. (2016). Fast Face Detection in Violent Video Scenes. *Electronic Notes in Theoretical Computer Science, 329*, 5-26. doi:10.1016/j.entcs.2016.12.002

Myles, D., Benoit-Barné, C., & Millerand, F. (2018). 'Not your personal army!' Investigating the organizing property of retributive vigilantism in a Reddit collective of websleuths. *Information, Communication & Society, 23*(3), 317-336. doi:10.1080/1369118x.2018.1502336

Neng, L. A. R., & Chambel, T. (2010). Get around 360° hypervideo. In Proceedings of the 14th International Academic MindTrek Conference: Envisioning Future Media Environments. *MindTrek'10*

119--122. doi:10.1145/1930488.1930512

Nguyen, C., Niu, Y., & Liu, F. (2014). Direct manipulation video navigation on touch screens. In Proceedings of the 16th international conference on Human-computer interaction with mobile devices & services (MobileHCI '14). *Association for Computing Machinery*, 273–282. doi:10.1145/2628363.2628365

Nhan, J., Huey, L., & Broll, R. (2015). Digilantism: An Analysis of Crowdsourcing and the Boston Marathon Bombings. *British Journal of Criminology*. doi:10.1093/bjc/azv118

Nilsen, L. G., Hafstad, G. S., Staksrud, E., & Dyb, G. (2018). Five reasons for using social media among young terror survivors: Results from the Utøya study. *Computers in Human Behavior, 84*, 285-294. doi:10.1016/j.chb.2018.03.006
Pantumsinchai, P. (2018). Armchair detectives and the social construction of falsehoods: an actor–network approach. *Information, Communication & Society, 21*(5), 761-778. doi:10.1080/1369118x.2018.1428654
Patel, R., & Davidson, B. (2019). *Forskningsmetodikens grunder - Att planera, genomföra och rapportera en undersökning* (5 ed.). Lund: Studentlitteratur.
Pouyanfar, S., Sadiq, S., Yan, Y., Tian, H., Tao, Y., Reyes, M. P., . . . Iyengar, S. S. (2019). A Survey on Deep Learning. *ACM Computing Surveys, 51*(5), 1-36. doi:10.1145/3234150
Pouyanfar, S., Tao, Y., Tian, H., Chen, S.-C., & Shyu, M.-L. (2018). Multimodal deep learning based on multiple correspondence analysis for disaster management. *World Wide Web, 22*(5), 1893-1911. doi:10.1007/s11280-018-0636-4
Pouyanfar, S., Yang, Y., Chen, S.-C., Shyu, M.-L., & Iyengar, S. S. (2018). Multimedia Big Data Analytics. *ACM Computing Surveys, 51*(1), 1-34. doi:10.1145/3150226
Rienecker, L., & Stray Jørgensen, P. (2017). *Att skriva en bra uppsats* (4 ed.). Stockholm: Liber.
Rubin, V. L. (2019). Disinformation and misinformation triangle. *Journal of Documentation, 75*(5), 1013-1034. doi:10.1108/jd-12-2018-0209
Schoeffmann, K., Hudelist, M. A., & Huber, J. (2015). Video Interaction Tools. *ACM Computing Surveys, 48*(1), 1-34. doi:10.1145/2808796
Simon, T., Goldberg, A., & Adini, B. (2015). Socializing in emergencies—A review of the use of social media in emergency situations. *International Journal of Information Management, 35*(5), 609-619. doi:10.1016/j.ijinfomgt.2015.07.001
Sreenu, G., & Saleem Durai, M. A. (2019). Intelligent video surveillance: a review through deep learning techniques for crowd analysis. *Journal of Big Data, 6*(1). doi:10.1186/s40537-019-0212-5
Starbird, K., Maddock, J., Orand, M., Achterman, P., & Mason, R. M. (2014). Rumors, False Flags, and Digital Vigilantes: Misinformation on Twitter After the 2013 Boston Marathon Bombing. *IConference*, 9. doi:10.9776/14308
Stolterman, E., & Wiberg, M. (2010). Concept-Driven Interaction Design Research. *Human–Computer Interaction, 25*(2), 95-118. doi:10.1080/07370020903586696
Techopedia. (2021, 2021). Labeled Data. Retrieved from https://www.techopedia.com/definition/33695/labeled-data
Thoresen, S., Jensen, T. K., & Dyb, G. (2014). Media participation and mental health in terrorist attack survivors. *J Trauma Stress, 27*(6), 639-646. doi:10.1002/jts.21971
Tian, H., Tao, Y., Pouyanfar, S., Chen, S.-C., & Shyu, M.-L. (2018). Multimodal deep representation learning for video classification. *World Wide Web, 22*(3), 1325-1341. doi:10.1007/s11280-018-0548-3
Trottier, D. (2016). Digital Vigilantism as Weaponisation of Visibility. *Philosophy & Technology, 30*(1), 55-72. doi:10.1007/s13347-016-0216-4
Vosoughi, S., Roy, D., & Aral, S. (2018). The spread of true and false news online. *Science, 359*(6380), 1146-1151. doi:10.1126/science.aap9559
Wang, B., & Zhuang, J. (2018). Rumor response, debunking response, and decision makings of misinformed Twitter users during disasters. *Natural Hazards, 93*(3), 1145-1162. doi:10.1007/s11069-018-3344-6
Wang, J., & Xu, Z. (2016). Spatio-temporal texture modelling for real-time crowd anomaly detection. *Computer Vision and Image Understanding, 144*, 177-187. doi:10.1016/j.cviu.2015.08.010
Wijnants, M., Rovelo, G., Quax, P., & Lamotte, W. (2016). WanderCouch. *Multimedia Tools and Applications, 76*(4), 5721-5753. doi:10.1007/s11042-016-3888-y
Wu, X., & Montgomery, M. (2019). Witnessing in crisis contexts in the social media age: the case of the 2015 Tianjin blasts on Weibo. *Media, Culture & Society, 42*(5), 675-691. doi:10.1177/0163443719855300

Yardley, E., Lynes, A. G. T., Wilson, D., & Kelly, E. (2016). What's the deal with 'websleuthing'? News media representations of amateur detectives in networked spaces. *Crime, Media, Culture: An International Journal, 14*(1), 81-109. doi:10.1177/1741659016674045

Ye, X., Peng, Z., & Guo, H. (2010). Spatio-Temporal Data Model and Spatio-Temporal Databases. *Temporal Information Processing Technology and Its Application*. doi:10.1007/978-3-642-14959-7_5

Zeitzoff, T. (2017). How Social Media Is Changing Conflict. *Journal of Conflict Resolution, 61*(9), 1970-1991. doi:10.1177/0022002717721392

Zubiaga, A., Liakata, M., Procter, R., Wong Sak Hoi, G., & Tolmie, P. (2016). Analysing How People Orient to and Spread Rumours in Social Media by Looking at Conversational Threads. *PLoS One, 11*(3), e0150989. doi:10.1371/journal.pone.0150989

INTERVIEW QUESTIONS - APPENDIX 1

1) In your opinion, what are some of the benefits/pros of this app/prototype/artefact?

2) How would the map affect your attitude towards the app and the events portrayed ?

3) How would the timeline affect your attitude towards the app and the events portrayed?

4) How would changing camera angles within the video affect your attitude towards the app, and the events portrayed?

5) How would the possibility to listen to audio from other camera angles affect your attitude towards the app and the events portrayed?

6) How would the possibility to interact directly with objects in the video affect your attitude towards this app and the events portrayed?

7) Overall, How do you experience the app?

8) Can you see any potential ethical implications with this app?

9) How do you feel about consuming this type of video material in comparison to other social media platforms?

10) Overall, Is there anything you feel is missing in this app?

11) Overall, Is there anything that you believe is redundant in f this app?

12) What other contexts/ areas could this application be applied to other than this type of events?